In [ ]:
```
 EXPERIMENT 2 – NUMPY OPERATIONS
PIN NO.: 241UDS0097

 AIM :-
To study and practice NumPy operations including array creation, attributes,
indexing, slicing, broadcasting, arithmetic, statistical operations,
concatenation, reshaping, sorting, splitting, and retrieving selected data.
```

In [ ]:
```
## Step 1 – Install and Import NumPy
If NumPy is not installed, run `%pip install numpy` in a separate code cell
first.
```

In [1]:
```
import numpy as np
```

In [ ]:
```
## Step 2 – Create Arrays for Three Energy Sources
Each array contains the output for Week 1 through Week 4.
```

In [2]:
```
solar_output_241UDS0097 = np.array([420, 455, 440, 480])
wind_output_241UDS0097 = np.array([365, 390, 410, 385])
hydro_output_241UDS0097 = np.array([510, 495, 530, 550])

solar_output_241UDS0097, wind_output_241UDS0097, hydro_output_241UDS0097
```

Out[2]:
```
(array([420, 455, 440, 480]),
 array([365, 390, 410, 385]),
 array([510, 495, 530, 550]))
```

In [ ]:
```
## Step 3 – Check Array Properties
```

In [3]:
```
solar_output_241UDS0097.ndim
```

Out[3]:
```
1
```

In [5]:
```
wind_output_241UDS0097.shape
```

Out[5]:
```
(4,)
```

In [6]:
```
hydro_output_241UDS0097.size
```

Out[6]:
```
4
```

In [7]:
```
solar_output_241UDS0097.dtype
```

Out[7]:
```
dtype('int64')
```

In [8]:
```
solar_output_241UDS0097.itemsize, wind_output_241UDS0097.itemsize,
hydro_output_241UDS0097.itemsize
```

Out[8]:
```
(8, 8, 8)
```

In [ ]:
```
## Step 4 — Retrieve Data Using Indexing and Slicing
```

In [9]:
```
hydro_output_241UDS0097[0]
```

Out[9]:
```
np.int64(510)
```

In [10]:
```
np.unique(wind_output_241UDS0097)
```

Out[10]:
```
array([365, 385, 390, 410])
```

In [11]:
```
solar_output_241UDS0097[0:2]
```

Out[11]:
```
array([420, 455])
```

In [12]:
```
wind_output_241UDS0097[2:]
```

Out[12]:
```
array([410, 385])
```

In [13]:
```
hydro_output_241UDS0097[-3:]
```

Out[13]:
```
array([495, 530, 550])
```

In [ ]:
```
## Step 5 — Assign New Values Using Broadcasting
Copy the Solar array, append two new weekly values, then replace those last
two values.
```

In [14]:
```
solar_output_new_241UDS0097 = solar_output_241UDS0097.copy()
solar_output_new_241UDS0097
```

Out[14]:
```
array([420, 455, 440, 480])
```

In [15]:
```
solar_output_new_241UDS0097 = np.append(solar_output_new_241UDS0097, [500,
515])
solar_output_new_241UDS0097
```

Out[15]:
```
array([420, 455, 440, 480, 500, 515])
```

In [16]:
```
solar_output_new_241UDS0097[4:6] = [505, 520]
solar_output_new_241UDS0097
```

Out[16]:
```
array([420, 455, 440, 480, 505, 520])
```

In [ ]:
```
## Step 6 — Arithmetic Operations
```

In [17]:
```
hydro_output_241UDS0097
```

Out[17]:
```
array([510, 495, 530, 550])
```

In [18]:
```
hydro_output_241UDS0097 + 25
```

Out[18]:
```
array([535, 520, 555, 575])
```

In [19]:
```
hydro_output_241UDS0097[3] - 15
```

Out[19]:
```
np.int64(535)
```

In [20]:
```
hydro_output_241UDS0097[0] * 2
```

Out[20]:
```
np.int64(1020)
```

In [21]:
```
hydro_output_241UDS0097[1:3] / 2
```

Out[21]:
```
array([247.5, 265. ])
```

In [ ]:
```
## Step 7 — Statistical Operations
```

In [22]:
```
solar_output_241UDS0097, wind_output_241UDS0097, hydro_output_241UDS0097
```

Out[22]:
```
(array([420, 455, 440, 480]),
 array([365, 390, 410, 385]),
 array([510, 495, 530, 550]))
```

In [23]:
```
np.sum(solar_output_241UDS0097)
```

Out[23]:
```
np.int64(1795)
```

In [25]:
```
np.max(solar_output_241UDS0097), np.min(solar_output_241UDS0097)
```

Out[25]:
```
(np.int64(480), np.int64(420))
```

In [26]:
```
print(
    np.mean(solar_output_241UDS0097),
    np.mean(wind_output_241UDS0097),
    np.mean(hydro_output_241UDS0097)
)
```

```
448.75 387.5 521.25
```

In [ ]:
```
## Step 8 — Combine All Energy-Source Data
```

In [27]:
```
solar_output_241UDS0097.ndim, wind_output_241UDS0097.ndim,
hydro_output_241UDS0097.ndim
```

Out[27]:
```
(1, 1, 1)
```

In [28]:
```
all_energy_output_241UDS0097 = np.concatenate([
    solar_output_241UDS0097,
    wind_output_241UDS0097,
    hydro_output_241UDS0097
])
all_energy_output_241UDS0097
```

Out[28]:
```
array([420, 455, 440, 480, 365, 390, 410, 385, 510, 495, 530, 550])
```

In [30]:
```
np.sum(all_energy_output_241UDS0097)
```

Out[30]:
```
np.int64(5430)
```

In [ ]:
```
## Step 9 — Reshape into a 3 × 4 Matrix
```

In [31]:
```
all_energy_output_241UDS0097.ndim, all_energy_output_241UDS0097.shape
```

Out[31]:
```
(1, (12,))
```

In [32]:
```
all_energy_output_241UDS0097 = all_energy_output_241UDS0097.reshape(3, 4)
all_energy_output_241UDS0097
```

Out[32]:
```
array([[420, 455, 440, 480],
       [365, 390, 410, 385],
       [510, 495, 530, 550]])
```

In [33]:
```
all_energy_output_241UDS0097.ndim, all_energy_output_241UDS0097.shape
```

Out[33]:
```
(2, (3, 4))
```

In [ ]:
```
## Step 10 — Sort the Array
```

In [34]:
```
all_energy_output_241UDS0097
```

Out[34]:
```
array([[420, 455, 440, 480],
       [365, 390, 410, 385],
       [510, 495, 530, 550]])
```

In [35]:
```
all_energy_output_241UDS0097[0]
```

Out[35]:
```
array([420, 455, 440, 480])
```

In [36]:
```
np.sort(all_energy_output_241UDS0097, axis=1)
```

Out[36]:
```
array([[420, 440, 455, 480],
       [365, 385, 390, 410],
       [495, 510, 530, 550]])
```

In [46]:
```
np.sort(all_energy_output_241UDS0097, axis=0)
```

Out[46]:
```
array([[365, 390, 410, 385],
       [420, 455, 440, 480],
       [510, 495, 530, 550]])
```

In [ ]:
```
## Step 11 — Split the Array
```

In [53]:
```
split_solar_241UDS0097, split_wind_241UDS0097, split_hydro_241UDS0097 =
np.split(
    all_energy_output_241UDS0097,
    [1, 2],
    axis=0
)
```

In [54]:
```
split_solar_241UDS0097
```

Out[54]:
```
array([[420, 455, 440, 480]])
```

In [49]:
```
split_wind_241UDS0097
```

Out[49]:
```
array([[365, 390, 410, 385]])
```

In [50]:
```
split_hydro_241UDS0097
```

Out[50]:
```
array([[510, 495, 530, 550]])
```

In [51]:
```
## Step 12 — Retrieve Week 2 Output of All Energy Sources
```

In [55]:
```
_2_output_241UDS0097 = all_energy_output_241UDS0097[:3, 1:2]
print(week_2_output_241UDS0097)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[55], line 2
      1 _2_output_241UDS0097 = all_energy_output_241UDS0097[:3, 1:2]
----> 2 print(week_2_output_241UDS0097)

NameError: name 'week_2_output_241UDS0097' is not defined
```

In [ ]:
```
 RESULT :-
Using NumPy, arrays were created and their properties inspected. Indexing,
slicing, value updates, arithmetic, statistical operations, concatenation,
reshaping, sorting, splitting, and selecting Week 2 values were performed on
the renewable-energy dataset.

**CSV note:** The CSV is provided as a dataset copy/reference. The
experiment operations intentionally follow the source PDF and create NumPy
arrays directly in Step 2.
```